

QCM Exam - Artificial Intelligence

Nicolas Caron

Instructions: There is only one good answer for each question. All questions are based on the course slides.

Each question is worth 1 point. There are no negative points.

First Name: _____ **Last Name:** _____

1. What is the main goal of Artificial Intelligence?
 - A. To replace humans in all physical jobs.
 - B. To design machines capable of performing tasks that usually require human intelligence.**
 - C. To build faster and more energy-efficient computers.
 - D. To solve complex mathematical equations symbolically.
2. Alan Turing invented artificial intelligence through the decryption of Enigma. (True or False)
 - A. True
 - B. False**
3. Which breakthrough launched the modern deep learning era?
 - A. The Turing Test (1950)
 - B. The Dartmouth workshop (1956)
 - C. Expert systems (1980s)
 - D. AlexNet / ImageNet (2012)**
4. Which statement is TRUE?
 - A. Machine Learning is broader than AI.
 - B. AI is broader than ML.**
 - C. AI and ML are exact synonyms.
 - D. Deep Learning is broader than AI.
5. Predicting whether an image contains a cat or a dog is an example of:
 - A. Regression
 - B. Clustering
 - C. Classification**
 - D. Natural Language Processing

6. Which task belongs to unsupervised learning?
- A. Regression
 - B. Classification
 - C. Clustering**
 - D. Game AI
7. In linear regression $y = wx + b$, what do we aim to optimize?
- A. x and y
 - B. w and b**
 - C. y only
 - D. x only
8. What is the purpose of a loss function?
- A. Measure model error.**
 - B. Add non-linearity.
 - C. Initialize the parameters.
 - D. Normalize the dataset.
9. If the gradient of a weight is positive, what will happen to the value of that weight during gradient descent?
- A. Decrease**
 - B. Increase
 - C. Remain fixed
 - D. Explode
10. According to the course, what is one major limitation of AI?
- A. It can only learn from unstructured text.
 - B. Data dependency (No data, no learning).**
 - C. It perfectly generalizes to any unseen situation.
 - D. It operates without any computational cost.
11. A larger model will always perform better than a smaller model.
- A. True
 - B. False**
12. What happens if we stack two linear layers without an activation function?
- A. It creates a highly non-linear model.
 - B. It is mathematically equivalent to a single linear layer.**
 - C. It automatically normalizes the data.
 - D. It becomes a Convolutional Neural Network.
13. Why are convolutional layers better suited for images?
- A. They flatten the image into a 1D vector instantly.
 - B. They keep spatial context through local filters.**

- C. They ignore color channels.
 - D. They require more parameters than a linear layer.
14. A batch size of 100 is used on a dataset of 1000 samples. How many gradient updates are performed in one epoch?
- A. 1
 - B. 10**
 - C. 100
 - D. 1000
15. In `nn.Linear(64, 256)`, what does 256 represent?
- A. Input features
 - B. Output neurons**
 - C. Batch size
 - D. Number of layers
16. What is the main role of a ReLU activation?
- A. Non linearity**
 - B. Normalize the output
 - C. Add spatial context
 - D. Decrease the loss
17. What does MaxPool2D keep from each local window?
- A. The average value
 - B. The minimum value
 - C. Select maximum value**
 - D. The sum of all values
18. What is the main purpose of Batch Normalization?
- A. Make the model smaller
 - B. Add non-linearity
 - C. Stabilize learning**
 - D. Perform classification
19. LayerNorm normalizes each feature across samples, while BatchNorm normalizes all features within a single sample.
- A. True
 - B. False**
20. How many trainable parameters does `nn.Linear(10, 5)` contain?
- A. 15
 - B. 50
 - C. 55**
 - D. 60

21. How many trainable parameters does this model contain?

```
import torch.nn as nn
model = nn.Sequential(
    nn.Linear(10, 20),
    nn.ReLU()
)
```

A. 200

B. 220

C. 240

D. 20

22. What is the Mean Squared Error (MSE) typically used for?

A. To add non-linearity to the model

B. As an activation function for classification

C. As a loss function for regression tasks

D. To normalize the dataset

23. Which of the following is an example of regression?

A. Predicting a continuous value like temperature

B. Identifying a cat or a dog in an image

C. Grouping customers into segments

D. Choosing the best move in a chess game

24. Grouping similar data points without predefined labels is known as:

A. Supervised learning

B. Unsupervised learning

C. Game AI

D. Reinforcement learning

25. What is the definition of an "Epoch" in neural network training?

A. A subset of the training data used for one gradient update

B. The initialization phase of the weights

C. The model has seen the entire training dataset once

D. A single forward pass through the network

26. Which of the following is a property of a good loss function?

A. Can be negative

B. Zero for perfect prediction

C. Unstable gradients

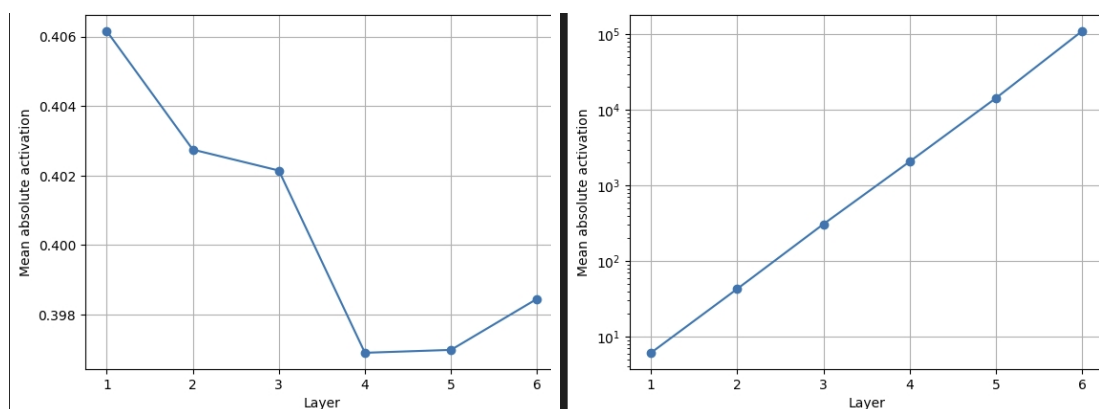
D. Non-differentiable

27. If the gradient is 0 during training, what happens to the parameters?

A. They increase

B. They decrease

- C. They remain fixed**
- D. They are reset to zero
28. Does the initial state (random initialization) of the weights prevent a simple linear regression from converging?
- A. Yes, it will completely fail to converge.
- B. No, gradient descent will move towards the optimum regardless.**
- C. Only if the learning rate is 0.
- D. Yes, it requires exact initialization.
29. In Deep learning models, what happens if the learning rate (η) is too large?
- A. Training is very slow but safe.
- B. There is a risk that the parameters overshoot the optimal values.**
- C. The model perfectly fits the training data immediately.
- D. It automatically reduces the number of epochs.
30. Why do we often standardize (normalize) features before training?
- A. Because unbalanced scales cause unstable and unbalanced gradients.**
- B. Because it increases the computational cost of the model.
- C. Because it acts as an activation function.
- D. Because it reduces the number of parameters in the model.
31. In modern language models, we often seek to:
- A. Predict the next word knowing the context.**
- B. Exploit the local spatial structure of an image.
- C. Group similar data points without predefined labels.
- D. Minimize the size of the convolution kernel.
32. What is the difference between the two plots?



- A. Normalized activation vs Exploding activation.**
- B. High learning rate vs Low learning rate.
- C. Overfitting vs Underfitting.
- D. Linear layers vs Convolutional layers.

33. What is the shape of the weight tensor for a PyTorch `nn.Conv2d(in_channels=3, out_channels=16, kernel_size=5)` layer?
- A. **[16, 3, 5, 5]**
 - B. [3, 16, 5, 5]
 - C. [16, 5, 5]
 - D. [3, 5, 5]
34. A standard scalar linear regression $\hat{y} = wx + b$ can be viewed as:
- A. A full convolutional layer.
 - B. **One artificial neuron without an activation function.**
 - C. A deep neural network.
 - D. An attention layer.
35. In PyTorch, what is the shape of the weight matrix W for the layer `nn.Linear(64, 256)`?
- A. [64, 256]
 - B. **[256, 64]**
 - C. [256]
 - D. [64]
36. What is the mathematical formula for the ReLU activation function?
- A. **$\max(0, x)$**
 - B. $\min(0, x)$
 - C. $1/(1 + e^{-x})$
 - D. x^2
37. An image with RGB channels is mathematically represented as a:
- A. 1D vector
 - B. 2D matrix
 - C. **3D tensor**
 - D. 4D tensor
38. What does the `kernel_size` define in a Convolutional layer?
- A. **The spatial size of the local filter window.**
 - B. The number of output channels.
 - C. The amount of padding added to the borders.
 - D. The size of the batch.
39. Mathematically performing a cognitive task is similar to:
- A. **Approximating a complex mathematical function.**
 - B. Solving a simple linear equation.
 - C. Memorizing a database perfectly.
 - D. Generating random numbers.
40. Why is MaxPool used in Convolutional Neural Networks?

- A. To add new learnable parameters.
 - B. To increase the spatial resolution of the image.
 - C. To reduce computation, memory usage, and improve robustness to small translations.**
 - D. To replace the ReLU activation.
41. Which normalization method normalizes features within a single example (across its features)?
- A. BatchNorm
 - B. LayerNorm**
 - C. MinMaxNorm
 - D. Z-Score
42. Which architecture was introduced in 2017 and is based on attention mechanisms?
- A. Expert Systems
 - B. ResNet
 - C. Transformer**
 - D. AlexNet
43. How is a convolutional filter applied to an image?
- A. Only once on the entire image
 - B. Repeatedly across different spatial locations**
 - C. Independently on each output class
 - D. After the loss function is computed
44. How many parameters are in a Convolutional layer with `in_channels=2`, `out_channels=1`, and a `kernel_size=3x3`?
- A. 18
 - B. 19**
 - C. 9
 - D. 36
45. What is the main role of gradient descent?
- A. To find the minimum of the loss function by updating the model's parameters.**
 - B. To normalize the dataset before training.
 - C. To calculate the accuracy of the model on the test set.
 - D. To initialize the parameters with random values.
46. Which of the following is the equation of the standardization (where μ is the mean and σ is the standard deviation)?
- A. $z = \frac{x - \mu}{\sigma}$**
 - B. $z = \frac{x - x_{min}}{x_{max} - x_{min}}$
 - C. $z = x - \mu$
 - D. $z = \frac{x}{\sigma}$

47. What is the gradient descent update rule for a parameter W ?

- A. $W \leftarrow W + \eta \frac{\partial \mathcal{L}}{\partial W}$
- B. $W \leftarrow W - \eta \frac{\partial \mathcal{L}}{\partial W}$**
- C. $W \leftarrow W \times \frac{\partial \mathcal{L}}{\partial W}$
- D. $W \leftarrow \eta \frac{\partial \mathcal{L}}{\partial W}$

48. Consider the following PyTorch model. Is this model optimal for an image classification problem?

```
import torch.nn as nn

model = nn.Sequential(
    nn.Conv2d(3, 16, kernel_size=3),
    nn.BatchNorm2d(16),
    nn.Conv2d(16, 32, kernel_size=3),
    nn.BatchNorm2d(32),
    nn.Flatten(),
    nn.Linear(32 * 28 * 28, 10)
)
```

A. No, it lacks an activation function.

- B. No, there is no point adding a linear layer after a convolution, it just increases the size of the model.
 - C. No, because a CNN cannot be used for image classification without LayerNorm.
 - D. Yes, this is a standard and optimal Convolutional Neural Network architecture.
49. Which of the following statements about linear layers vs CNNs on images is true based on the course?
- A. Linear layers always achieve higher accuracy than CNNs on images.
 - B. CNNs obtain better accuracy with fewer parameters because they exploit the spatial structure.**
 - C. Linear layers preserve the 2D spatial context of pixels perfectly.
 - D. CNNs cannot be combined with BatchNorm.